

Module 05: Action — Video Games and Controls

Learning Objectives

1. Understand that **Action** in a video game happens through a **Controller**.
2. Learn about **Latency** (The speed of the signal).
3. Know that video games are "Feedback Loops" (You act -> Game reacts -> You react).

Introduction: The Magic Button

When you press "A" on your controller, Mario jumps. It feels instant. But it is a journey. Finger -> Button -> Wire -> Console -> TV -> Eyes. If any part of that chain is slow, you die in the game. This is the Perception-Action Loop at light speed.

Key Concepts

1. The Interface (The Bridge)

A controller is a bridge between your hands and the computer code.

- Buttons = Triggers.
- Joysticks = Direction vectors.
- Touchscreens = X,Y coordinates.

2. Latency (Lag)

Have you ever played a game that felt "sticky"? That is **Latency**. The delay between your brain saying "Jump!" and Mario jumping. Low Latency = Good responsiveness. High Latency = Frustration (Game Over).

3. Feedback (Rumble)

How do you know you got hit?

- Screen flashes red (Visual).
- Sound effect (Auditory).
- Controller shakes (Haptic). The game "touches" you back!

Activities

Activity 1: The Human Controller

Partner A is the Player (holding a fake controller). Partner B is the Character (standing in front). A presses "A" -> B jumps. A presses "Left" -> B steps left. Be fast! If B is slow, say "Lag!"

Activity 2: The Mirror Game

Stand face to face. Try to move your hands exactly at the same time. Who is leading? Who is following? In a perfect loop, action and perception happen together.

Summary

Video games are the best way to practice Active Inference. You predict, you act, you update.

References

- *Press Start* by Thomas Flintham
- *Trapped in a Video Game* by Dustin Brady